Spike Train Analysis

Pierre Le Merre

CNRS, Centre de Recherche en Neuroscience de Lyon
Dept. of Neuroscience, Karolinska Institute

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Plan

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Spike rates and intervals

Waveforms and cell types

Connectivity and Cross correlograms

Population coding and embedding

Decoding

Conclusion

Position of the problem

- Preprocessing is done (spike sorting, histology, behavior,...)
- We have unit spike trains and other timestamps stored in some format...
-

Position of the problem

- Preprocessing is done (spike sorting, histology, behavior,...)
- We have unit spike trains and other timestamps stored in some format...
- We have a research question.



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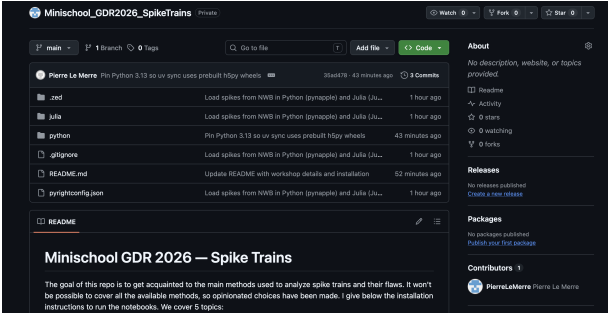
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Go to ⇒
`https://github.com/samuelgarcia/school_neuralnet_paris_2026`
folder `notebooks_spiketrain_analysis`



Download the data - Proposed framework



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Buccino et al, 2022

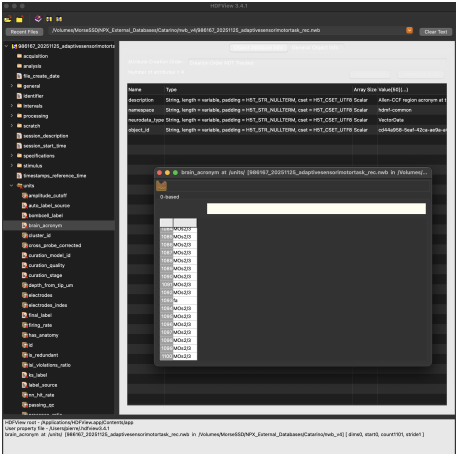
https://spikeinterface.readthedocs.io/en/stable/ https://nwb.org/ https://dandiarchive.org/

Download the data

link 2 data

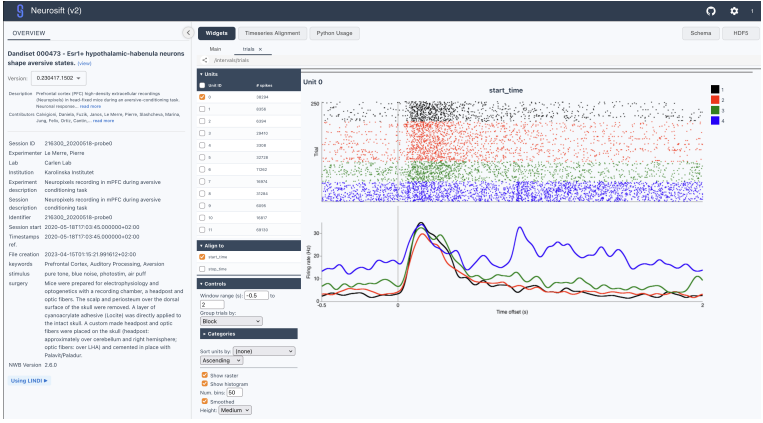
Explore NWB data - HDFView

https://github.com/HDFGroup/hdfview/releases/tag/v3.5.0

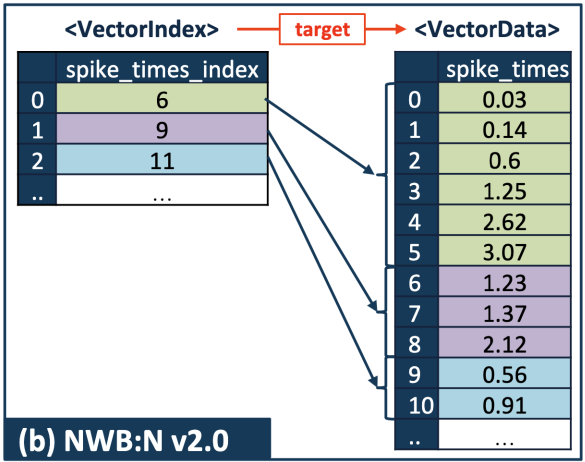


Explore NWB data - Neurosift

https://neurosift.app/



Explore NWB data - where are my spike times?



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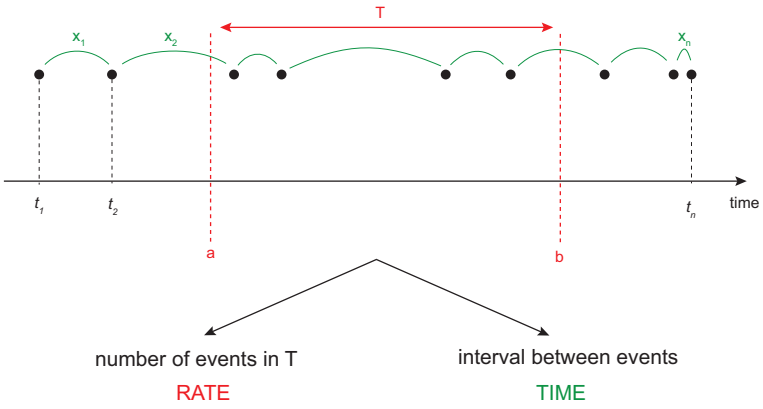
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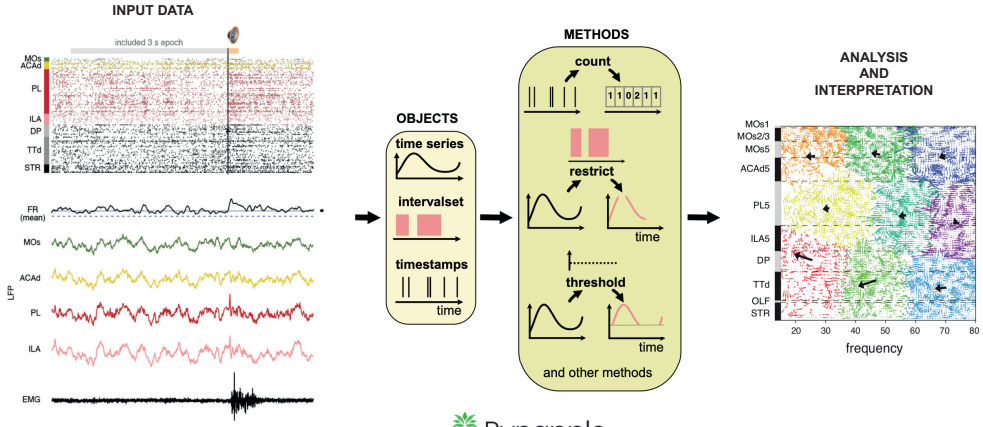
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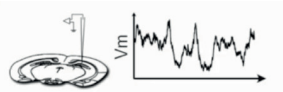
Spike trains are Point processes



Pynapple


INPUT DATA
included 3 s epoch
MOs
ACAd
PL
ILA
DP
TTd
STR
FR
(mean)
MOs
ACAd
PL
ILA
LFP
EMG
OBJECTS
time series
intervalset
timestamps
time
METHODS
count
1 1 0 2 1 1
restrict
time
threshold
time
and other methods
ANALYSIS
AND
INTERPRETATION
MOs1
MOs2/3
MOs5
ACAd5
PL5
ILA5
DP
TTd
OLF
STR
20
30
40
50
60
70
80
frequency
Pynapple

Pynapple objects



Tsd: 1-dimension

```
In [11]: tsd = nap.Tsd(t=t, d=d)
```



TsdFrame: 2-dimensions

```
In [20]: tsdframe = nap.TsdFrame(t=t, d=d,
    ...:                 columns = ['x', 'y'])
```



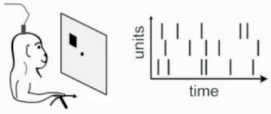
TsdTensor: n-dimensions

```
In [34]: tsdtensor = nap.TsdTensor(t=t, d=d)
```



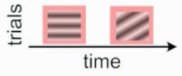
Ts: Timestamps

```
In [6]: nap.Ts(t)
```



TsGroup: group of timestamps

```
In [18]: nap.TsGroup(data=data)
```



IntervalSet: set of epochs

```
In [23]: nap.IntervalSet(
    ...:     start=start,
    ...:     end = end)
```

Hands on session

01-SpikeRate and SpikeIntervals.ipynb

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